

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination (New)

University Theory Examination November 2017

PH215 - Pharmacology-I

Date: 20/11/2017, Monday

Time: 10.00 AM to 1.00 PM

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat sketches wherever necessary.
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SECTION – I

Q.1 Attempt all questions. Each question is of one mark.

20

1. Which of the following statements best describes pharmacokinetics?
 - a) The study of how drugs reach their target in the body and how the level of a drug in the blood are affected by various factors
 - b) The study of how drugs can be designed using molecular modelling based on a drug's pharmacophore
 - c) The study of how a drug interacts with its target binding site at the molecular level
 - d) The study of which functional groups are important in binding a drug to its target binding site and the identification of a pharmacophore
2. Select the correct option for active transport of drug across the biological membrane?
 - a) Transport of drugs through a membrane by means of diffusion
 - b) Transport without energy consumption
 - c) Engulf of drug by a cell membrane with a new vesicle formation
 - d) Transport against concentration gradient
3. Which one of the following statements is false regarding the blood-brain barrier?
 - a) The walls of the capillaries supplying the brain have tight fitting cells making it difficult for polar drugs to leave the capillaries.
 - b) The capillaries in the brain have a fatty coating making it more difficult for drugs to enter the brain.
 - c) The walls of the capillaries supplying the brain are made up of several layers of cells, which act as a barrier to the release of drugs.
 - d) Hydrophobic drugs pass through the blood brain barrier more easily than hydrophilic drugs.
4. Which route of drug administration is most likely lead to the first-pass effect?
 - a) Sublingual
 - b) Oral
 - c) Intravenous
 - d) Intramuscular

5. Parenteral administration:
 - a) Cannot be used with unconsciousness patients
 - b) Generally results in a less accurate dosage than oral administration
 - c) Usually produces a more rapid response than oral administration
 - d) Is too slow for emergency use
6. Which of the following drugs has both alpha-1-selective and beta-blocking effects?
 - a) Labetalol
 - b) Betaxolol
 - c) Propranolol
 - d) Timolol
7. A teratogenic action is:
 - a) Toxic action on the liver
 - b) Toxic action on fetal malformation
 - c) Toxic action on blood system
 - d) Toxic action on kidneys
8. What does "affinity" mean?
 - a) A measure of how tightly a drug binds to plasma proteins
 - b) A measure of how tightly a drug binds to a receptor
 - c) A measure of inhibiting potency of a drug
 - d) A measure of bioavailability of a drug
9. A competitive antagonist is a substance that:
 - a) Interacts with receptors and produces submaximal effect
 - b) Binds to the same receptor site and progressively inhibits the agonist response
 - c) Binds to the nonspecific sites of tissue
 - d) Binds to one receptor subtype as an agonist and to another as an antagonist
10. The term "bioavailability" means
 - a) Plasma protein binding degree of substance
 - b) Permeability through the brain-blood barrier
 - c) Fraction of an uncharged drug reaching the systemic circulation following any route administration
 - d) Amount of a substance in urine relative to the initial dose
11. Tolerance and drug resistance can be consequence of:
 - a) Change in receptor and exhaustion of mediators
 - b) Increased receptor sensitivity
 - c) Decrease metabolic degradation
 - d) Activation of drug after hepatic first pass
12. Conjugation of a drug includes the following EXCEPT:
 - a) Glucuronidation
 - b) Sulfate formation
 - c) Hydrolysis
 - d) Methylation
13. In Cobra bite which is the primary treatment provided?
 - a) Combination of Noadrenaline and Adrenaline
 - b) Combination of Neostigmine and Atropine
 - c) Histamine antagonist
 - d) Combination of Rivastigmine and donepezil

14. Which of the following effects is related to direct beta1-adrenoreceptor stimulation?
 - a) Bronchodilation
 - b) Vasodilatation
 - c) Tachycardia
 - d) Bradycardia
15. Acetylcholine is not a specific neurotransmitter at:
 - a) Sympathetic ganglia
 - b) Sympathetic postganglionic nerve endings
 - c) Parasympathetic ganglia
 - d) Parasympathetic postganglionic nerve endings
16. Which of the following drugs is a nonselective beta receptor antagonist?
 - a) Metoprolol
 - b) Atenolol
 - c) Propranolol
 - d) Acebutolol
17. Indicate the location of M2 cholinoreceptor type:
 - a) Heart
 - b) Glands
 - c) Smooth muscle
 - d) Endothelium
18. Select a muscarinic receptor-blocking drug:
 - a) Scopolamine
 - b) Pilocarpine
 - c) Trimethaphan
 - d) Pilocarpine
19. The primary mechanism of action of local anesthetics is:
 - a) Activation of ligand-gated potassium channels
 - b) Blockade of voltage-gated sodium channels
 - c) Stimulation of voltage-gated N-type calcium channels
 - d) Blockade of the GABA-gated chloride channels
20. Subtype-selective alpha 1 receptor antagonists such as tamsulosin, terazosin, alfuzosin are efficacious in:
 - a) Hyperthyroidism
 - b) Cardiac arrhythmias
 - c) Benign prostatic hyperplasia (BPH)
 - d) Asthma

SECTION – II

Q.2 Attempt any TWO of the following

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| A Describe the detail mechanism of actions of Parasympathomimetic drugs | 05 |
| B Write a detail Pharmacological action of Atropine. | 05 |
| C Enlist the therapeutic uses of Parasympathomimetics and Parasympatholytic drugs. | 05 |

Q.3 Attempt any TWO of the following

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| A Study the Phenomenon of Dale's vasomotor reversal. Give the therapeutic uses of Sympathomimetic drugs. | 05 |
| B Classify Sympatholytic drugs in detail with examples. | 05 |
| C Study the Pharmacological actions and therapeutic uses of Beta adrenergic blocking agents. | 05 |

SECTION – III

Q 4 Attempt any FOUR of the following

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| A | Write a brief note on Local Anaesthetics | 05 |
| B | Write a note on factors affecting drug absorption. | 05 |
| C | Describe in brief about G-protein coupled receptors (GPCR). | 05 |
| D | Provide a detail note on oxidation reaction of drug metabolism. | 05 |
| E | Write down advantages and disadvantages of oral route of drug administration. | 05 |
| F | Write a note on Drug receptor interactions. | 05 |

Q 5 Attempt any FOUR of the following

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| A | Write a brief note on Neuromuscular blocking agents | 05 |
| B | Write a note on plasma protein binding of drugs. | 05 |
| C | Write a note on drug antagonism. | 05 |
| D | Write a detail note on bioavailability of drug. | 05 |
| E | Discuss the concept of drug interactions in detail. | 05 |
| F | Give a short note on adverse effects of drugs. | 05 |
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